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PS C:\Users\Chayce\Documents\CollegeFinalSemester\MachineLearning\HW5\HW5\code>
python main.py --task train --run_name larger_embd --data_split simple --n_layer
2 --n_head 2 --n_embd 32 --max_epochs 60 --batch_size 32 --num_workers 8 --
learning_rate 4e-4 --max_len 128 --seed 44 --grad_norm_clip 1.0
The file './tokenizer/simple_vocab.json' exists. Loading tokenizer.
{'<pad>': 0, '<s>': 1, '</s>': 2, '<unk>': 3, 'I_TURN_RIGHT': 4, 'I_JUMP': 5,
'I_WALK': 6, 'I_TURN_LEFT': 7, 'I_RUN': 8, 'I_LOOK': 9, 'jump': 10, 'opposite':
11, 'right': 12, 'twice': 13, 'and': 14, 'turn': 15, 'thrice': 16, 'run': 17,
'left': 18, 'after': 19, 'walk': 20, 'around': 21, 'look': 22}
train dataset size: 15055
val dataset size: 1673
loading model
total params: 31104
C:
\Users\Chayce\Documents\CollegeFinalSemester\MachineLearning\HW5\HW5\code\traine
r.py:79: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated.
Please use `torch.amp.GradScaler('cuda', args...)` instead.
    scaler = GradScaler()
C:\Users\Chayce\AppData\Roaming\Python\Python312\site-
packages\torch\amp\grad_scaler.py:132: UserWarning: torch.cuda.amp.GradScaler is
enabled, but CUDA is not available. Disabling.
    warnings.warn(
epoch 1 iter 0: train loss 0.00000. lr 0.0000e+00: 0%|
| 0/471 [00:00<?, ?it/s]C
:
\Users\Chayce\Documents\CollegeFinalSemester\MachineLearning\HW5\HW5\code\traine
r.py:102: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated.
Please use `torch.amp.autocast('cuda', args...)` instead.
    with torch.cuda.amp.autocast():
C:\Users\Chayce\AppData\Roaming\Python\Python312\site-
packages\torch\amp\autocast_mode.py:266: UserWarning: User provided device_type
of 'cuda', but CUDA is not available. Disabling
    warnings.warn(
epoch 1 iter 470: train loss 0.65824. lr 3.9978e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:18<00:00, 25.63it/s]
test loss: %f 0.7838543732211275
epoch_valid_loss: 0.7838543732211275, epoch_train_loss: 1.5066138002270093,
epoch: 1
Saving at epoch 1:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.7354297637939453 train_step: 500, learning_rate:
0.0003997489407867487
27/471 [00:01<00:16, 26.92it/s]
epoch 2 iter 470: train loss 0.74326. lr 3.9902e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:18<00:00, 25.40it/s]
test loss: %f 0.5566759030773955
epoch_valid_loss: 0.5566759030773955, epoch_train_loss: 0.719478131099871,
epoch: 2
Saving at epoch 2:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.6420586109161377 train_step: 1000, learning_rate:
0.0003988910329278014
57/471 [00:02<00:20, 20.24it/s]
epoch 3 iter 470: train loss 0.71608. lr 3.9773e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:21<00:00, 21.45it/s]
test loss: %f 0.4274629365723088
epoch_valid_loss: 0.4274629365723088, epoch_train_loss: 0.6042985571924036,
epoch: 3
Saving at epoch 3:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt

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step_train_loss: 0.4917634129524231 train_step: 1500, learning_rate:
0.00039742625820294794
87/471 [00:03<00:15, 25.06it/s]
epoch 4 iter 470: train loss 0.45630. lr 3.9590e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:20<00:00, 23.55it/s]
test loss: %f 0.370943916856118
epoch_valid_loss: 0.370943916856118, epoch_train_loss: 0.5229842621682809,
epoch: 4
Saving at epoch 4:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.4801320731639862 train_step: 2000, learning_rate:
0.00039535908601049877
114/471 [00:04<00:14, 25.04it/s]
epoch 5 iter 470: train loss 0.34099. lr 3.9354e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:20<00:00, 23.51it/s]
test loss: %f 0.31859068713098204
epoch_valid_loss: 0.31859068713098204, epoch_train_loss: 0.476143039842067,
epoch: 5
Saving at epoch 5:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.48285624384880066 train_step: 2500, learning_rate:
0.0003926958238158596
144/471 [00:05<00:13, 24.86it/s]
epoch 6 iter 470: train loss 0.31897. lr 3.9065e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:19<00:00, 23.56it/s]
test loss: %f 0.29870119100471715
epoch_valid_loss: 0.29870119100471715, epoch_train_loss: 0.4360149778631336,
epoch: 6
Saving at epoch 6:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.375230610370636 train_step: 3000, learning_rate:
0.00038944459790585885
174/471 [00:07<00:12, 24.71it/s]
epoch 7 iter 470: train loss 0.50770. lr 3.8725e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:19<00:00, 23.66it/s]
test loss: %f 0.26539155672181325
epoch_valid_loss: 0.26539155672181325, epoch_train_loss: 0.4089020625421196,
epoch: 7
Saving at epoch 7:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.45112037658691406 train_step: 3500, learning_rate:
0.00038561532859338987
201/471 [00:08<00:10, 24.77it/s]
epoch 8 iter 470: train loss 0.40517. lr 3.8334e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:20<00:00, 23.44it/s]
test loss: %f 0.24587076340081557
epoch_valid_loss: 0.24587076340081557, epoch_train_loss: 0.38638426037604134,
epoch: 8
Saving at epoch 8:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.38050517439842224 train_step: 4000, learning_rate:
0.00038121969994802686
231/471 [00:09<00:09, 24.81it/s]
epoch 9 iter 470: train loss 0.38800. lr 3.7894e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:20<00:00, 23.44it/s]
test loss: %f 0.23249736400145404
epoch_valid_loss: 0.23249736400145404, epoch_train_loss: 0.36877868731057317,
epoch: 9
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Saving at epoch 9:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.35138756036758423 train_step: 4500, learning_rate:
0.00037627112671667753
261/471 [00:10<00:08, 24.72it/s]
epoch 10 iter 470: train loss 0.36480. lr 3.7405e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████| 471/471 [00:20<00:00, 23.54it/s]
test loss: %f 0.22877704422428924
epoch_valid_loss: 0.22877704422428924, epoch_train_loss: 0.3538267340060252,
epoch: 10
Saving at epoch 10:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.2775886058807373 train_step: 5000, learning_rate:
0.0003707847005411132
288/471 [00:12<00:07, 24.02it/s]
epoch 11 iter 470: train loss 0.26535. lr 3.6869e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████| 471/471 [00:21<00:00, 22.12it/s]
test loss: %f 0.21124960622697506
epoch_valid_loss: 0.21124960622697506, epoch_train_loss: 0.3461167829431546,
epoch: 11
Saving at epoch 11:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.3134496212005615 train_step: 5500, learning_rate:
0.0003647771665180489
318/471 [00:12<00:06, 25.27it/s]
epoch 12 iter 470: train loss 0.36725. lr 3.6287e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████| 471/471 [00:20<00:00, 23.52it/s]
test loss: %f 0.19767017268909598
epoch_valid_loss: 0.19767017268909598, epoch_train_loss: 0.33348178683192864,
epoch: 12
Saving at epoch 12:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.33189669251441956 train_step: 6000, learning_rate:
0.0003582668618277934
348/471 [00:14<00:05, 24.46it/s]
epoch 13 iter 470: train loss 0.45351. lr 3.5661e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████| 471/471 [00:20<00:00, 23.51it/s]
test loss: %f 0.2017869010286511
epoch_valid_loss: 0.2017869010286511, epoch_train_loss: 0.3227157938252589,
epoch: 13
step_train_loss: 0.34425103664398193 train_step: 6500, learning_rate:
0.00035127364537228863
375/471 [00:15<00:03, 25.05it/s]
epoch 14 iter 470: train loss 0.28231. lr 3.4993e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████| 471/471 [00:20<00:00, 23.13it/s]
test loss: %f 0.19928030500996788
epoch_valid_loss: 0.19928030500996788, epoch_train_loss: 0.31819171798583556,
epoch: 14
step_train_loss: 0.2687333822250366 train_step: 7000, learning_rate:
0.00034381884763814557
405/471 [00:16<00:02, 24.84it/s]
epoch 15 iter 470: train loss 0.26846. lr 3.4284e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████| 471/471 [00:20<00:00, 23.45it/s]
test loss: %f 0.1781913936138153
epoch_valid_loss: 0.1781913936138153, epoch_train_loss: 0.3069761506501277,
epoch: 15
Saving at epoch 15:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
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step_train_loss: 0.24002060294151306 train_step: 7500, learning_rate:
0.00033592522177779913
435/471 [00:17<00:01, 25.38it/s]
epoch 16 iter 470: train loss 0.26693. lr 3.3536e-04: 100%|
| 471/471 [00:19<00:00, 23.57it/s]
test loss: %f 0.1688416565083108
epoch_valid_loss: 0.1688416565083108, epoch_train_loss: 0.3030763874641143,
epoch: 16
Saving at epoch 16:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.33472153544425964 train_step: 8000, learning_rate:
0.0003276168616470794
462/471 [00:18<00:00, 24.47it/s]
epoch 17 iter 470: train loss 0.32173. lr 3.2752e-04: 100%|
| 471/471 [00:19<00:00, 23.76it/s]
test loss: %f 0.1656728412182826
epoch_valid_loss: 0.1656728412182826, epoch_train_loss: 0.2959886586843276,
epoch: 17
Saving at epoch 17:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
epoch 18 iter 470: train loss 0.31847. lr 3.1934e-04: 100%|
| 471/471 [00:19<00:00, 23.63it/s]
test loss: %f 0.16166169604040542
epoch_valid_loss: 0.16166169604040542, epoch_train_loss: 0.2897684528569507,
epoch: 18
Saving at epoch 18:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.30285415053367615 train_step: 8500, learning_rate:
0.00031892856370618823
21/471 [00:00<00:18, 24.72it/s]
epoch 19 iter 470: train loss 0.27670. lr 3.1083e-04: 100%|
| 471/471 [00:20<00:00, 23.49it/s]
test loss: %f 0.14739648224610202
epoch_valid_loss: 0.14739648224610202, epoch_train_loss: 0.283568841351825,
epoch: 19
Saving at epoch 19:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.32823681831359863 train_step: 9000, learning_rate:
0.00030986831410617134
51/471 [00:02<00:16, 25.45it/s]
epoch 20 iter 470: train loss 0.25896. lr 3.0202e-04: 100%|
| 471/471 [00:19<00:00, 23.77it/s]
test loss: %f 0.1509816380885412
epoch_valid_loss: 0.1509816380885412, epoch_train_loss: 0.2771512512948103,
epoch: 20
step_train_loss: 0.3176887035369873 train_step: 9500, learning_rate:
0.0003004728463160256
78/471 [00:03<00:15, 25.14it/s]
epoch 21 iter 470: train loss 0.34168. lr 2.9293e-04: 100%|
| 471/471 [00:19<00:00, 23.71it/s]
test loss: %f 0.1409824143322009
epoch_valid_loss: 0.1409824143322009, epoch_train_loss: 0.27599166188143875,
epoch: 21
Saving at epoch 21:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.2652914226055145 train_step: 10000, learning_rate:
0.00029077081075159177
108/471 [00:04<00:14, 25.35it/s]
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epoch 22 iter 470: train loss 0.19188. lr 2.8359e-04: 100%|

| 471/471 [00:19<00:00, 23.64it/s]
test loss: %f 0.1355355405863726
epoch_valid_loss: 0.1355355405863726, epoch_train_loss: 0.2701152573535397,
epoch: 22

Saving at epoch 22:

./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt

step_train_loss: 0.29896679520606995 train_step: 10500, learning_rate:
0.00028079179083144917

138/471 [00:05<00:13, 25.28it/s]

epoch 23 iter 470: train loss 0.26773. lr 2.7403e-04: 100%|

| 471/471 [00:19<00:00, 23.79it/s]
test loss: %f 0.1366487533416388
epoch_valid_loss: 0.1366487533416388, epoch_train_loss: 0.26299557507417765,
epoch: 23

step_train_loss: 0.26063108444213867 train_step: 11000, learning_rate:
0.00027056627435953035

165/471 [00:06<00:12, 25.12it/s]

epoch 24 iter 470: train loss 0.22544. lr 2.6427e-04: 100%|

| 471/471 [00:19<00:00, 23.63it/s]
test loss: %f 0.13767487555742264
epoch_valid_loss: 0.13767487555742264, epoch_train_loss: 0.2624236667259526,
epoch: 24

step_train_loss: 0.23218317329883575 train_step: 11500, learning_rate:
0.00026012544254159104

195/471 [00:07<00:11, 24.66it/s]

epoch 25 iter 470: train loss 0.26044. lr 2.5433e-04: 100%|

| 471/471 [00:20<00:00, 23.39it/s]
test loss: %f 0.1241919662592546
epoch_valid_loss: 0.1241919662592546, epoch_train_loss: 0.2555981753611514,
epoch: 25

Saving at epoch 25:

./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt

step_train_loss: 0.3031231462955475 train_step: 12000, learning_rate:
0.00024950115296592005

225/471 [00:09<00:10, 23.04it/s]

epoch 26 iter 470: train loss 0.24106. lr 2.4424e-04: 100%|

| 471/471 [00:21<00:00, 21.45it/s]
test loss: %f 0.13283372529835072
epoch_valid_loss: 0.13283372529835072, epoch_train_loss: 0.25222310879427917,
epoch: 26

step_train_loss: 0.21144451200962067 train_step: 12500, learning_rate:
0.00023872582299629312

252/471 [00:11<00:10, 20.44it/s]

epoch 27 iter 470: train loss 0.21286. lr 2.3404e-04: 100%|

| 471/471 [00:23<00:00, 19.90it/s]
test loss: %f 0.11690860649324814
epoch_valid_loss: 0.11690860649324814, epoch_train_loss: 0.24780889899487707,
epoch: 27

Saving at epoch 27:

./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt

step_train_loss: 0.26755404472351074 train_step: 13000, learning_rate:
0.0002278323092971486

281/471 [00:14<00:09, 20.75it/s]

epoch 28 iter 470: train loss 0.28787. lr 2.2374e-04: 100%|

| 471/471 [00:24<00:00, 19.05it/s]

test loss: %f 0.11557567541329365

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epoch_valid_loss: 0.11557567541329365, epoch_train_loss: 0.24520440029490526,
epoch: 28
Saving at epoch 28:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.2687057554721832 train_step: 13500, learning_rate:
0.00021685389362485693
312/471 [00:12<00:06, 24.76it/s]
epoch 29 iter 470: train loss 0.25365. lr 2.1338e-04: 100%|
| 471/471 [00:20<00:00, 23.55it/s]
test loss: %f 0.11782114559187079
epoch_valid_loss: 0.11782114559187079, epoch_train_loss: 0.2409071913760179,
epoch: 29
step_train_loss: 0.21408264338970184 train_step: 14000, learning_rate:
0.00020582405250931256
339/471 [00:13<00:05, 24.80it/s]
epoch 30 iter 470: train loss 0.22149. lr 2.0298e-04: 100%|
| 471/471 [00:20<00:00, 23.44it/s]
test loss: %f 0.10338476398643458
epoch_valid_loss: 0.10338476398643458, epoch_train_loss: 0.23762312186624848,
epoch: 30
Saving at epoch 30:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.2802988886833191 train_step: 14500, learning_rate:
0.0001947764189876604
369/471 [00:14<00:04, 24.77it/s]
epoch 31 iter 470: train loss 0.47621. lr 1.9258e-04: 100%|
| 471/471 [00:20<00:00, 22.82it/s]
test loss: %f 0.10736270389466915
epoch_valid_loss: 0.10736270389466915, epoch_train_loss: 0.2346042301624444,
epoch: 31
step_train_loss: 0.34888210892677307 train_step: 15000, learning_rate:
0.00018374474568164296
398/471 [00:17<00:03, 20.28it/s]
epoch 32 iter 470: train loss 0.12906. lr 1.8219e-04: 100%|
| 471/471 [00:22<00:00, 21.27it/s]
test loss: %f 0.10702526569366455
epoch_valid_loss: 0.10702526569366455, epoch_train_loss: 0.23164819642869783,
epoch: 32
step_train_loss: 0.2732061743736267 train_step: 15500, learning_rate:
0.00017276267122068255
427/471 [00:18<00:01, 25.08it/s]
epoch 33 iter 470: train loss 0.18417. lr 1.7185e-04: 100%|
| 471/471 [00:21<00:00, 22.06it/s]
test loss: %f 0.09668626397285822
epoch_valid_loss: 0.09668626397285822, epoch_train_loss: 0.22788450915975428,
epoch: 33
Saving at epoch 33:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.230534166097641 train_step: 16000, learning_rate:
0.000161863704659693
455/471 [00:20<00:00, 23.67it/s]
epoch 34 iter 470: train loss 0.20446. lr 1.6159e-04: 100%|
| 471/471 [00:21<00:00, 21.81it/s]
test loss: %f 0.10417015695909285
epoch_valid_loss: 0.10417015695909285, epoch_train_loss: 0.2257847943503386,
epoch: 34
epoch 35 iter 470: train loss 0.21463. lr 1.5144e-04: 100%|
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██████████ | 471/471 [00:23<00:00, 20.41it/s]
test loss: %f 0.10143149700367225
epoch_valid_loss: 0.10143149700367225, epoch_train_loss: 0.22559344819143826,
epoch: 35
step_train_loss: 0.22200751304626465 train_step: 16500, learning_rate:
0.0001510924808426786
15/471 [00:00<00:18, 24.61it/s]
epoch 36 iter 470: train loss 0.16825. lr 1.4141e-04: 100%|
```

```
██████████ | 471/471 [00:20<00:00, 23.39it/s]
test loss: %f 0.0954191378967942
epoch_valid_loss: 0.0954191378967942, epoch_train_loss: 0.2187380107661468,
epoch: 36
Saving at epoch 36:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.17119884490966797 train_step: 17000, learning_rate:
0.00014045898638681664
42/471 [00:01<00:17, 24.48it/s]
epoch 37 iter 470: train loss 0.24885. lr 1.3154e-04: 100%|
```

```
██████████ | 471/471 [00:21<00:00, 21.91it/s]
test loss: %f 0.09276721120442984
epoch_valid_loss: 0.09276721120442984, epoch_train_loss: 0.22020854824667524,
epoch: 37
Saving at epoch 37:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.21784691512584686 train_step: 17500, learning_rate:
0.00013000710454508208
72/471 [00:03<00:16, 23.64it/s]
epoch 38 iter 470: train loss 0.21069. lr 1.2186e-04: 100%|
```

```
██████████ | 471/471 [00:21<00:00, 22.40it/s]
test loss: %f 0.09237924176004697
epoch_valid_loss: 0.09237924176004697, epoch_train_loss: 0.21850993672500496,
epoch: 38
Saving at epoch 38:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.23790450394153595 train_step: 18000, learning_rate:
0.00011976883009889161
102/471 [00:04<00:16, 23.04it/s]
epoch 39 iter 470: train loss 0.15180. lr 1.1239e-04: 100%|
```

```
██████████ | 471/471 [00:21<00:00, 22.35it/s]
test loss: %f 0.08547713980078697
epoch_valid_loss: 0.08547713980078697, epoch_train_loss: 0.21313448310404573,
epoch: 39
Saving at epoch 39:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.2159312218427658 train_step: 18500, learning_rate:
0.00010977536101143186
129/471 [00:05<00:14, 23.43it/s]
epoch 40 iter 470: train loss 0.27533. lr 1.0316e-04: 100%|
```

```
██████████ | 471/471 [00:20<00:00, 23.06it/s]
test loss: %f 0.08970611816588438
epoch_valid_loss: 0.08970611816588438, epoch_train_loss: 0.21236880164743735,
epoch: 40
step_train_loss: 0.23785115778446198 train_step: 19000, learning_rate:
0.00010005718985545179
160/471 [00:07<00:12, 24.45it/s]
epoch 41 iter 470: train loss 0.31251. lr 9.4187e-05: 100%|
```

```
██████████ | 471/471 [00:20<00:00, 22.87it/s]
test loss: %f 0.0843005746884166
```

```
epoch_valid_loss: 0.0843005746884166, epoch_train_loss: 0.21182116002413878,
epoch: 41
Saving at epoch 41:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.18857336044311523 train_step: 19500, learning_rate:
9.064396920088103e-05
189/471 [00:07<00:11, 24.98it/s]
epoch 42 iter 470: train loss 0.18905. lr 8.5503e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:20<00:00, 23.49it/s]
test loss: %f 0.0846186656294004
epoch_valid_loss: 0.0846186656294004, epoch_train_loss: 0.20991238926373215,
epoch: 42
step_train_loss: 0.22939036786556244 train_step: 20000, learning_rate:
8.156442113742624e-05
216/471 [00:08<00:10, 24.70it/s]
epoch 43 iter 470: train loss 0.25856. lr 7.7128e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:19<00:00, 23.61it/s]
test loss: %f 0.08460797657662968
epoch_valid_loss: 0.08460797657662968, epoch_train_loss: 0.2084665689340569,
epoch: 43
step_train_loss: 0.20715853571891785 train_step: 20500, learning_rate:
7.284624963629354e-05
246/471 [00:10<00:09, 24.86it/s]
epoch 44 iter 470: train loss 0.18426. lr 6.9086e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:19<00:00, 23.56it/s]
test loss: %f 0.08233931379498176
epoch_valid_loss: 0.08233931379498176, epoch_train_loss: 0.2051047541976228,
epoch: 44
Saving at epoch 44:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.12876571714878082 train_step: 21000, learning_rate:
6.451602398585763e-05
276/471 [00:11<00:08, 24.19it/s]
epoch 45 iter 470: train loss 0.24902. lr 6.1399e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:20<00:00, 23.11it/s]
test loss: %f 0.08120943880024946
epoch_valid_loss: 0.08120943880024946, epoch_train_loss: 0.20364089740436295,
epoch: 45
Saving at epoch 45:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.17092521488666534 train_step: 21500, learning_rate:
5.659922743246404e-05
303/471 [00:12<00:06, 24.29it/s]
epoch 46 iter 470: train loss 0.19222. lr 5.4087e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:20<00:00, 23.51it/s]
test loss: %f 0.076988862370545
epoch_valid_loss: 0.076988862370545, epoch_train_loss: 0.20347814739636302,
epoch: 46
Saving at epoch 46:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.19512352347373962 train_step: 22000, learning_rate:
4.911998248917342e-05
333/471 [00:13<00:05, 24.29it/s]
epoch 47 iter 470: train loss 0.17780. lr 4.7170e-05: 100%|
████████████████████████████████████████████████████████████████████████████████
████████████████████████████████████████████████████████████████████████████████
| 471/471 [00:19<00:00, 23.63it/s]
test loss: %f 0.07750994177921763
epoch_valid_loss: 0.07750994177921763, epoch_train_loss: 0.20256314833199648,
epoch: 47
```

```
step_train_loss: 0.13147054612636566 train_step: 22500, learning_rate:
4.210111020226544e-05
363/471 [00:15<00:04, 23.73it/s]
epoch 48 iter 470: train loss 0.15396. lr 4.0667e-05: 100%|
| 471/471 [00:21<00:00, 22.26it/s]
test loss: %f 0.07795514187441682
epoch_valid_loss: 0.07795514187441682, epoch_train_loss: 0.20124905883886252,
epoch: 48
step_train_loss: 0.18133243918418884 train_step: 23000, learning_rate:
4e-05
390/471 [00:16<00:03, 20.90it/s]
epoch 49 iter 470: train loss 0.22163. lr 4.0000e-05: 100%|
| 471/471 [00:21<00:00, 22.21it/s]
test loss: %f 0.08118383888647242
epoch_valid_loss: 0.08118383888647242, epoch_train_loss: 0.20248712498038185,
epoch: 49
step_train_loss: 0.17397800087928772 train_step: 23500, learning_rate:
4e-05
420/471 [00:17<00:02, 23.33it/s]
epoch 50 iter 470: train loss 0.21492. lr 4.0000e-05: 100%|
| 471/471 [00:21<00:00, 22.40it/s]
test loss: %f 0.07681508123312357
epoch_valid_loss: 0.07681508123312357, epoch_train_loss: 0.20081465961826836,
epoch: 50
Saving at epoch 50:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.18243128061294556 train_step: 24000, learning_rate:
4e-05
450/471 [00:18<00:00, 23.71it/s]
epoch 51 iter 470: train loss 0.25952. lr 4.0000e-05: 100%|
| 471/471 [00:20<00:00, 22.75it/s]
test loss: %f 0.07549081377263339
epoch_valid_loss: 0.07549081377263339, epoch_train_loss: 0.19851022780004834,
epoch: 51
Saving at epoch 51:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
epoch 52 iter 470: train loss 0.17380. lr 4.0000e-05: 100%|
| 471/471 [00:20<00:00, 23.03it/s]
test loss: %f 0.0747356365241532
epoch_valid_loss: 0.0747356365241532, epoch_train_loss: 0.19822134637528924,
epoch: 52
Saving at epoch 52:
./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt
step_train_loss: 0.2184169441461563 train_step: 24500, learning_rate: 4e-05
| 6/471 [00:00<00:19, 23.59it/s]
epoch 53 iter 470: train loss 0.24949. lr 4.0000e-05: 100%|
| 471/471 [00:21<00:00, 21.72it/s]
test loss: %f 0.07609610435254169
epoch_valid_loss: 0.07609610435254169, epoch_train_loss: 0.19623793343639678,
epoch: 53
step_train_loss: 0.22140410542488098 train_step: 25000, learning_rate: 4e-05
| 36/471 [00:01<00:18, 24.13it/s]
epoch 54 iter 470: train loss 0.19205. lr 4.0000e-05: 100%|
| 471/471 [00:20<00:00, 22.66it/s]
test loss: %f 0.07546393601399548
epoch_valid_loss: 0.07546393601399548, epoch_train_loss: 0.19922245690903592,
epoch: 54
```

step_train_loss: 0.22306528687477112 train_step: 25500, learning_rate: 4e-05
| 66/471 [00:02<00:16, 23.96it/s]

epoch 55 iter 470: train loss 0.26417. lr 4.0000e-05: 100%|

| 471/471 [00:20<00:00, 22.65it/s]

test loss: %f 0.07818265716141125

epoch_valid_loss: 0.07818265716141125, epoch_train_loss: 0.19846934423388443,
epoch: 55

step_train_loss: 0.1917940378189087 train_step: 26000, learning_rate: 4e-05

| 94/471 [00:04<00:16, 23.16it/s]

epoch 56 iter 470: train loss 0.24639. lr 4.0000e-05: 100%|

| 471/471 [00:21<00:00, 21.64it/s]

test loss: %f 0.07889651907783635

epoch_valid_loss: 0.07889651907783635, epoch_train_loss: 0.19815958814889256,
epoch: 56

step_train_loss: 0.21368414163589478 train_step: 26500, learning_rate:

4e-05

| 122/471 [00:05<00:15, 23.05it/s]

epoch 57 iter 470: train loss 0.14465. lr 4.0000e-05: 100%|

| 471/471 [00:21<00:00, 21.57it/s]

test loss: %f 0.07640211487997253

epoch_valid_loss: 0.07640211487997253, epoch_train_loss: 0.19602822563007885,
epoch: 57

step_train_loss: 0.22735927999019623 train_step: 27000, learning_rate:

4e-05

| 153/471 [00:06<00:13, 23.04it/s]

epoch 58 iter 470: train loss 0.19718. lr 4.0000e-05: 100%|

| 471/471 [00:21<00:00, 21.70it/s]

test loss: %f 0.07591710935504932

epoch_valid_loss: 0.07591710935504932, epoch_train_loss: 0.19713622026319463,
epoch: 58

step_train_loss: 0.2261740118265152 train_step: 27500, learning_rate:

4e-05

| 182/471 [00:08<00:12, 22.62it/s]

epoch 59 iter 470: train loss 0.17536. lr 4.0000e-05: 100%|

| 471/471 [00:21<00:00, 21.69it/s]

test loss: %f 0.07428785294013203

epoch_valid_loss: 0.07428785294013203, epoch_train_loss: 0.1961450825134913,
epoch: 59

Saving at epoch 59:

./cond_gpt/weights/larger_embd_simplesplit_2layer_2head_32embd_32bs.pt

step_train_loss: 0.21199221909046173 train_step: 28000, learning_rate:

4e-05

| 210/471 [00:09<00:11, 22.62it/s]

epoch 60 iter 470: train loss 0.14677. lr 4.0000e-05: 100%|

| 471/471 [00:21<00:00, 21.55it/s]

test loss: %f 0.07441298642529631

epoch_valid_loss: 0.07441298642529631, epoch_train_loss: 0.19507814562561152,
epoch: 60